MOTRIT-GPT

LLMs-based Interactive Tool for Multilayer Optical Thin-films Research

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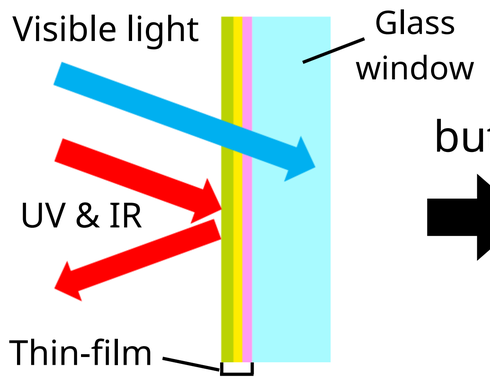
1. Introduction



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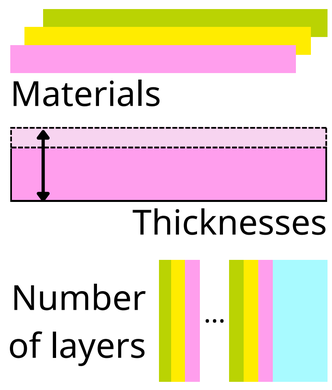
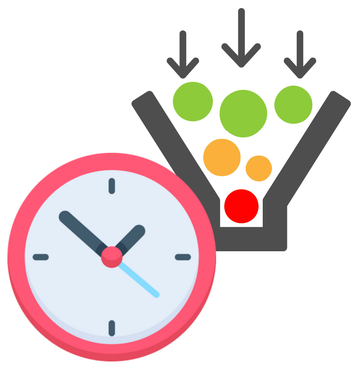


leads to



Thin-films have **crucial** applications (e.g. energy saving to **reduce carbon footprint**)

Thin-films require **many precise** parameters that are **hard to optimise** for applications

Significant **bottleneck**; Hours spent with **no optimal** thin-film configuration found

Aims

Large language models (LLMs) can converse well with humans.

Develop LLMs-based **MOTRIT-GPT** to:

a. **Automate** design tasks, enhancing **efficiency** in thin-film research.

b. Enable **quick** access to **expert** information by natural language queries.

2. Methodology

Autonomous Design

User input → Add context prompt to message → OpenAI's GPT-3.5 API → Response generated → Python APIs (`visualise_film`, `simulate_film`, `optimise_film`) → Results displayed

more paramaters needed

all data given

`visualise_film`
Draws thin film structure with Matplotlib

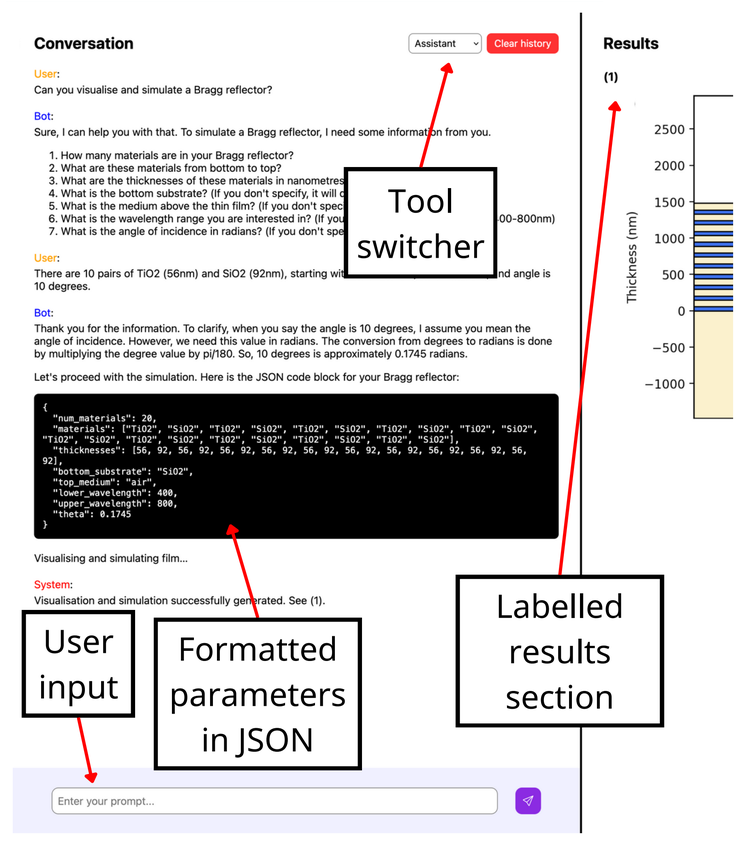
`simulate_film`
Predicts spectrum by Transfer Matrix Method

`optimise_film`
Local: Stochastic Gradient Descent (SGD) to find a good fit for the goal
Global: Select good initial structures to use SGD on

Knowledge Query

User input + Info database → OpenAI's Embeddings API → Similarity querying between embeddings to find relevant info in database → Combine info and question → OpenAI's GPT-3.5 API → Results displayed

Graphical User Interface



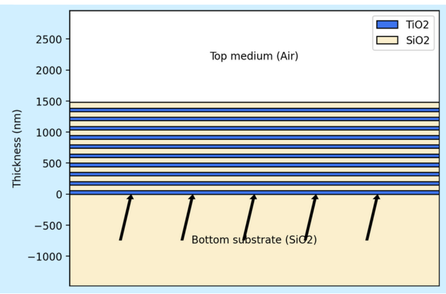
3. Results and Discussion

Autonomous Design

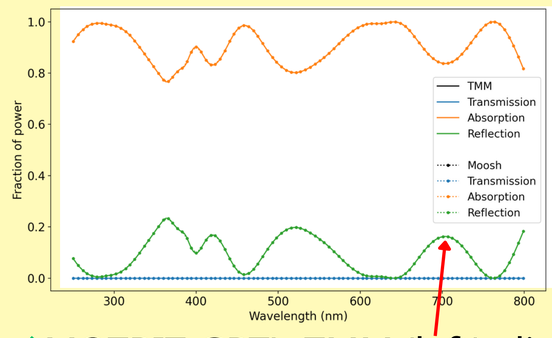
Visualisation

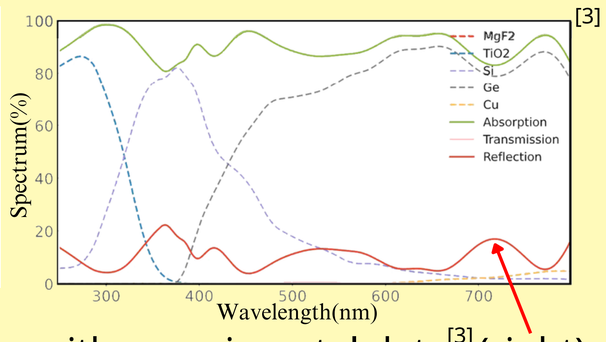
✓ Provides clarity in the structure

✓ Shows light direction and angle



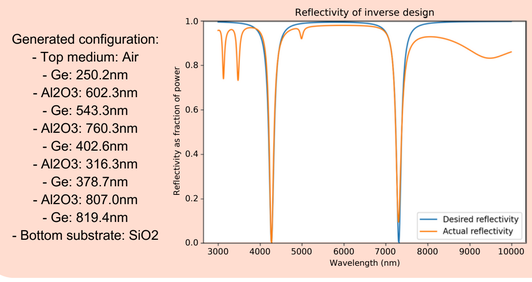
Simulation





✓ MOTRIT-GPT's TMM (left) align with experimental data[3] (right)

✓ MOTRIT-GPT's TMM (left) align with algorithmic data (left)



Local-based Inverse Design

✓ Provides control over materials

✓ Faster computation

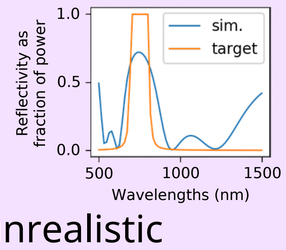
× Potentially lower accuracy as it finds local (not global) minima

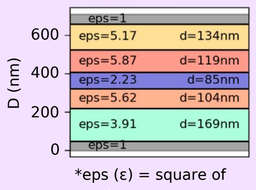
Global-based Inverse Design

✓ Yields near-global minima results, enhancing accuracy

× Longer computation time

× Suggested materials may be unrealistic





*eps (ε) = square of refractive index

Knowledge Query

Q: What is the working principle of TMM-Fast and how is it faster than its counterparts?

Bot:
The working principle of TMM-Fast is based on the Transfer Matrix Method (TMM), which is used to calculate the optical response of a thin film to an incident plane wave. TMM-Fast is a parallelized and re-implemented version of the existing TMM code, initially published by Byrnes. It implements the Abèles TMM in Python to calculate the optical properties of layered thin-film materials with different thicknesses.

TMM-Fast achieves faster computation compared to its counterparts through parallelization. It utilizes the thread-management package Dask to distribute the computation across multiple CPU cores. This parallelization significantly reduces the computational time, resulting in a speedup of approximately 100 times compared to the original implementation by Byrnes. Additionally, TMM-Fast reaches computational speeds comparable to state-of-the-art software, such as the stackrt function from Lumerical.

ChatGPT
As of my last knowledge update in January 2022, I don't have specific information about a technology or system called "TMM-Fast." It's possible that it's a term or concept that has emerged after my last update, or it could be a specific technology that wasn't widely known or documented up to that point.
If TMM-Fast is a new development or a niche technology that has emerged since then, I recommend checking the latest scientific literature, research papers, or official documentation for detailed information on its working principle and how it achieves speed improvements compared to its counterparts.

✓ MOTRIT-GPT (left) is more comprehensive than ChatGPT (right)

✓ MOTRIT-GPT avoids technical details for ease of understanding

Capability of MOTRIT-GPT

>90% of time saved by autonomous design

Many hours of time saved by knowledge query

4. Conclusion

MOTRIT-GPT **lowers barriers** for designers

MOTRIT-GPT **accelerates** design processes

Future work includes:

- integrate autonomous design and knowledge query tools
- customise GUI results
- enhance `optimise_film` (e.g. through DRL methods)

References
[1] Fan, L. *et al.* Thin-film neural networks for optical inverse problem. *Light: Advanced Manufacturing* **2**, 27 (2021).
[2] He, Z. *et al.* ChatEDA: A Large Language Model powered autonomous agent for EDA. Preprint at https://arxiv.org/abs/2308.10204 (2023).
[3] Jiang, A. & Osamu, Y. A reinforcement learning method for optical thin-film design. *IEICE Trans. Electron.* **E105–C**, 95–101 (2022).

All icons were taken from Flaticon. OpenAI logo was taken from https://openai.com. All images and graphs were self-drawn unless otherwise stated.